ITIL Group Reports Frontend — Production Deployment MOP

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Field	Value
Document reference	MOP-ITIL-FE-001
Application	ITIL Group Reports Automation — Next.js frontend
Target environment	Production (first deployment)
Deployment owner	Application team
Infrastructure model	Rootless Podman with Nginx ingress
Reference environment	QA
Status	🟡 Pending — Section 3 values not yet allocated by Infrastructure

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1. Purpose and scope

This document defines the production prerequisites and deployment method for the frontend.

Infrastructure prepares the host, service account, storage, network, DNS, TLS, and Nginx.

> ⚠️ **Warning:** Production-specific values that are not yet allocated are intentionally blank in Section 3. Infrastructure should complete and return that section **before** the deployment session.

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2. Target architecture

```
User browser
    |
    | HTTPS 443
    v
Nginx (production FQDN and TLS)
    |
    | HTTP to 127.0.0.1:3001
    v
Next.js frontend container (rootless Podman, port 3000)
    |
    | /api/* over Podman network "itil-net"
```

```
    v
FastAPI backend container (port 8000, not publicly exposed)
    |
    v
PostgreSQL container (port 5432, internal only)

```

> ℹ️ **Note:** Browser requests remain same-origin. The frontend sends requests to `/api/v1/*`; the Next.js server proxies them to the backend origin configured when the frontend release is built. Browser-to-backend CORS is therefore not required for the normal production path.

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3. Production values to be completed by Infrastructure

Item	Production value
Linux hostname	
Server IP	
Rootless service account	
Public frontend FQDN	
DNS change owner/reference	
TLS certificate owner/reference	
Nginx configuration owner	
Approved deployment workstation/source	
Approved container registry or image-transfer method	
Backup retention period	
Monitoring/alerting destination	

> ℹ️ **Note — backend origin:** The backend origin is **not** an Infrastructure-selected public URL. The application team will set it after the backend is deployed. When both containers use `itil-net`, the intended value is:

```
http://itil-backend-api:8000

```

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4. Infrastructure prerequisites

4.1 Host and service account

- Supported production Linux distribution.
- Rootless Podman 4.x or later for the dedicated service account.
- `loginctl enable-linger <service-account>` so containers survive logout and can start after host reboot.
- Subordinate UID/GID mappings configured where required.
- `bash`, `curl`, `tar`, `gzip`, `ca-certificates`, and an approved SSH/SFTP transfer mechanism.
- NTP enabled and timezone set to `Africa/Johannesburg`.
- SSH access from the approved deployment workstation.

Validation:

```
id
podman --version
podman info --format '{{.Host.Security.Rootless}}'
loginctl show-user "$USER" -p Linger

```

Expected values: the production service account, `true` for rootless, and `Linger=yes`.

4.2 Directories and permissions

Create the shared layout used by QA:

```
sudo mkdir -p /opt/itil-report-stack/scripts
sudo mkdir -p /opt/itil-report-stack/frontend/releases
sudo chown -R svc_itil_reports:svc_itil_reports /opt/itil-report-stack
sudo chmod 750 /opt/itil-report-stack

```

> ℹ️ **Note — shared host layout:** `/opt/itil-report-stack/scripts` is shared across both applications for common operational scripts. Each application owns its own `releases/` and `current` symlink under its own subdirectory (`frontend/` , `backend/`) so the two MOPs never write to the same path — see MOP-ITIL-REP-001 for the backend equivalent (`/opt/itil-report-stack/backend/...`).

Frontend releases use:

```
/opt/itil-report-stack/frontend/releases/<release-tag>/
/opt/itil-report-stack/frontend/current -> releases/<release-tag>/

```

> ℹ️ **Note:** Retain at least one previously verified frontend release after each upgrade.

4.3 Podman network

The application team creates the shared rootless bridge network if it does not exist:

```
podman network inspect itil-net >/dev/null 2>&1 ||
  podman network create itil-net

```

> ⚠️ **Warning:** The frontend, backend, and PostgreSQL containers must be created by the **same** rootless service account to use this network consistently.

4.4 Network and firewall

Direction	Port	Source/destination	Purpose
Inbound	TCP 22	Approved admin/deployment workstation	Deployment
Inbound	TCP 443	Approved user networks	Nginx HTTPS
Inbound	TCP 80	Approved user networks	Optional HTTPS redirect only

Host-local	TCP 3001	`127.0.0.1` only	Nginx to frontend
Container internal	TCP 8000	Frontend to backend on `itil-net`	API
Container internal	TCP 5432	Backend to PostgreSQL on `itil-net`	Database
Outbound	TCP 443	Approved registry	Image access, if used

> ⚠️ **Warning:** Ports `3001`, `8000`, and `5432` must **not** be exposed to user networks.

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5. Nginx requirement

Infrastructure should create a production server block using the completed FQDN and approved TLS certificate:

```
server {
    listen 80;
    server_name <production-frontend-fqdn>;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name <production-frontend-fqdn>;

    ssl_certificate     <approved-certificate-path>;
    ssl_certificate_key <approved-private-key-path>;

    client_max_body_size 20m;

    location / {
        proxy_pass http://127.0.0.1:3001;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_read_timeout 120s;
    }
}

```

Infrastructure may add approved security headers and corporate TLS standards.

> ⚠️ **Warning:** Do not cache `/api/*`, login responses, or authenticated HTML.

Validate before go-live:

```
sudo nginx -t
curl --fail --resolve <production-frontend-fqdn>:443:127.0.0.1 \
  https://<production-frontend-fqdn>/healthz

```

---

6. Release preparation

The release must:

- Be built from the approved source revision.
- Pass `npm run type-check`, `npm run lint`, and `npm run build`.
- Be built for Linux using the repository Dockerfile or an approved Linux CI runner.
- Use `NEXT_PUBLIC_API_URL` as an empty value so browser calls remain same-origin.
- Use the application-team-confirmed internal backend origin for `API_PROXY_URL`.
- Record the source revision, release tag, build time, and image/archive checksum in the change record.

The repository Dockerfile (`node:22-bookworm-slim`) is the authoritative frontend runtime. Production should not substitute `node:22-alpine` for a standalone bundle built on another operating system.

Example image build after the backend origin is confirmed:

```
podman build \
  --build-arg NEXT_PUBLIC_API_URL= \
  --build-arg API_PROXY_URL=http://itil-backend-api:8000 \
  --tag localhost/itil-frontend-ui:<release-tag> \
  .

```

> ⚠️ **Warning — confirm at runtime too:** `API_PROXY_URL` is set above as a build arg. Confirm whether the Next.js runtime (e.g. `next.config.js` rewrites) also reads this value from the **container's environment at startup**, not only at build time. If there's any doubt, also pass `--env API_PROXY_URL=http://itil-backend-api:8000` on `podman run` (Sections 7 and 9) — harmless if not needed, but prevents a silently broken proxy if it is.

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7. First production deployment

Run as the rootless production service account:

```
podman network inspect itil-net >/dev/null 2>&1 ||
  podman network create itil-net

podman run -d \
  --name itil-frontend-ui \
  --network itil-net \
  --restart unless-stopped \
  --publish 127.0.0.1:3001:3000 \
  --env NODE_ENV=production \
  --env PORT=3000 \
  --env HOSTNAME=0.0.0.0 \
  --env API_PROXY_URL=http://itil-backend-api:8000 \
  localhost/itil-frontend-ui:<release-tag>

```

The image contains a health check for `/healthz`. Confirm:

```
podman ps --filter name=itil-frontend-ui
podman inspect itil-frontend-ui \
  --format '{{.State.Status}} {{if .State.Health}}
{{.State.Health.Status}}{{end}}'
podman logs --tail 100 itil-frontend-ui
curl --fail http://127.0.0.1:3001/healthz

```

Then validate through Nginx:

```
curl --fail https://<production-frontend-fqdn>/healthz
curl --fail https://<production-frontend-fqdn>/api/v1/health

```

> ℹ️ **Note:** The second command proves the complete Nginx → frontend → backend path.

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8. Functional validation

During the approved deployment window, the application team will verify:

- Login using the temporary production Super Admin account.
- Forced password rotation.
- Workspace, SLA Monitor, Daily Report, and Raw Reports pages.
- Division and assignee-group data.
- Backend health and latest pipeline status.
- An authorised report generation test.
- No QA hostname, IP, credential, or data appears in production.
- Browser developer tools show `/api/v1/*` calls to the production frontend origin, with no CORS failures or proxy loops.

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9. Rollback

Before deployment, confirm the previous image exists:

```
podman image exists localhost/itil-frontend-ui:<previous-release-tag>

```

Rollback:

```
podman stop itil-frontend-ui
podman rm itil-frontend-ui

podman run -d \
  --name itil-frontend-ui \
  --network itil-net \
  --restart unless-stopped \
  --publish 127.0.0.1:3001:3000 \
  --env NODE_ENV=production \
  --env PORT=3000 \
  --env HOSTNAME=0.0.0.0 \
  --env API_PROXY_URL=http://itil-backend-api:8000 \
  localhost/itil-frontend-ui:<previous-release-tag>

curl --fail http://127.0.0.1:3001/healthz
curl --fail https://<production-frontend-fqdn>/healthz

```

> ℹ️ **Note:** The frontend has no database migration and can be rolled back independently, provided the selected frontend release remains API-compatible with the deployed backend.

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10. Monitoring and handover

Infrastructure should monitor:

- HTTPS availability of `https://<production-frontend-fqdn>/healthz`.
- Nginx 5xx response rate.
- Rootless container state/restarts.
- Host disk usage.
- TLS certificate expiry.

Application logs:

```
podman logs --since 30m itil-frontend-ui

```

> ℹ️ **Note:** The final operations handover must record the application owner, Infra support group, monitoring destination, escalation route, backup retention, production URL, and approved release tag.
